MIL-C-7188
10 May 1951
Superseding
AN-C-156
29 January 1946

MILITARY SPECIFICATION

COMPASSES, PILOT'S STANDBY, INSTALLATION OF

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers the installation of pilot's standby compasses.

2. APPLICABLE SPECIFICATIONS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.

Military

MIL-I-7028	Instruments and Instrument Boards; Aircraft Installation of

Air Force-Navy Aeronautical

AN-C-106	Compass Swinging; Aircraft
AN-C-107	Compasses; Installation of

(Copies of this specification and copies of other publications referenced or required for Government procurement, and the Index of Military Aeronautical (AN or MIL) Standards, may be obtained upon application to the Commanding General, Air Material Command, Wright-Patterson Air Force Base, Dayton, Ohio; or the Commanding Officer, U.S. Naval Air Station, Johnsville, Pennsylvania.)

3. REQUIREMENTS

3.1 General. The requirements specififed in Specification MIL-I-7028 are applicable as requirements of this specification.

3.1.1 In the event the requirements of this specification and the general specification conflict, the requirements of this specification shall govern.

3.2 Nonmagnetic Materials. All materials used in mounting the compass, such as brackets, screws, plate nuts, et cetera, shall be of nonmagnetic metal.

3.3 Location. The compass shall be located on the instrument panel or positioned on a bracket suitably insulated from vibration in some position off the instrument panel. The compass shall be positioned to be easily read by the pilot.

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3.3.1 Lubber Line. The compass shall be so installed that the vertical plan through the lubber line and the center of the pivot are parallel to the longitudinal axis of the airplane.

3.3.2 Leveling. When the compass is installed in a location other than the pilot's or copilot's instrument panel, it shall be so positioned that it is in a normal level position when the airplane is in normal flight attitude.

3.3.3 Labels. The compass installation conforming to this specification shall be clearly labeled with a suitable placard as follows:

"TO BE USED AS A STANDBY COMPASS ONLY."

3.4 Vibration Insulation. The compass mounting shall be vibration insulated in accordance with Specification MIL-I-7028.

3.5 Wiring. When the compass is equipped for individual lighting, the wire to the compass shall be a twisted two-wire system. The installation shall be acceptable to the Procuring Service.

3.5.1 Rheostat. A rheostat shall be conveniently located, in order that the pilot may control the intensity of lighting.

3.6 Performance. After installation, the compass shall satisfy the requirements specified in Section 4.

3.7 Demagnetization of Parts. Demagnetization of parts after assembling into the airplane shall be performed only with the specific approval of the Procuring Service.

3.8 Workmanship. Workmanship shall be in accordance with high-grade manufacturing practice in installations of this type and shall meet the approval of the Procuring Service.

4. SAMPLING, INSPECTION AND TEST PROCEDURES.

4.1 General. All the tests required for the testing of compass installations are classified as Inspection tests,for which necessary sampling techniques and methods of testing are specified in this section. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Procuring Service. The contractor shall make available test reports, for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.2 Test Conditions. All tests shall be subject to the test conditions specified herein.

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4.2.1 Standard Test Compass. The standard test compass shall be a direct-indicating compass which has previously been calibrated at each 5-degree magnetic heading so that it shall be capable of indicating the angle between the vertical plane containing the horizontal component of the earth's magnetic field and a vertical plane which contains the lubber's line within 1 degree,and shall be provided with a standard universal type compensator.

4.2.2 Installation of Standard Test Compass. Unless otherwise specified, the standard compass shall be installed, with the compensator removed, on a suitable bracket at the location at which the investigation is to be made and shall be installed in such a way that the plane through the lubber's line and the pivot point is parallel to the longitudinal axis of the airplane.

4.2.3 Tapping. The standard test compass shall be tapped lightly before all readings are taken, unless the engine is running.

4.2.4 Observer. The observer shall posses no magentic materials on his person such as screw drivers, pocket knives and other tools, mechanical pencils, wrist watches, bracelets, eye glasses that have magnetic frames, officer's caps, badges, shop buttons, et cetera.

4.2.5 Magnetic Heading. The magnetic heading of the airplane on the ground and in flight shall be determined in accordance with the methods specified in specification AN-C-106.

4.2.6 Local Field Strength. The changes of compass reading due to magnetic materials and electric currents, specified in this specification, apply only to locations at which the strength of the horizontal component of the earth's magnetic field (airplane removed) is 0.18 oersted. When tests are accomplished at locations where the strength of the horizontal component of the earth's field differs from 0.18 oersted, the allowable changes specified herein, shall be corrected as specified in table II of Specification AN-C-107.

4.2.7 Records. Permanent records shall be kept of all the data which are obtained during the tests.

4.2.8 Readings. All readings on the ground and in the air shall be taken to the nearest degree. When noting changes of compass reading, both magnitude and direction of the change shall be recorded with an increase of reading taken as positive and a decrease of reading take as negative.

4.3 Individual Tests. Each installation shall be inspected carefully to determine compliance with this specification with respect to workmanship, installation, and demagnetization.

4.4 Sampling Tests. One prototype, 1 of the first 10 production airplanes of any design, and such additional airplanes as may be specified by the Procuring Service, shall be selected by the inspector for subjection to the tests specified herein.

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4.4.1 Effect on Climb and Glide, Ground Swing. The airplane shall be placed on and east or west heading in an attitude of horizontal flight and in an attitude of 5 degress tail down. The compass shall be read in both attitudes with no electrical circuits on. In each attitude the circuits to be tested as specified under tests headed, "Effect of Variable Circuits" and "Effects of Continuously Operated Electrical Circuits," shall be energized and the compass reading noted. The change in the compass reading shall not exceed 3 degrees. All remaining tests, with the exception of the Effect of Engine test, shall be performed with the airplane in the horzontal flight attitude.

4.4.1.1 This test need not be repeated on other headings. If a nonmagnetic jack or dolly is unavailable, one having magnetic materials may be used provided the following requirements is met. The jack (or dolly) shall be moved in a circle about the compass in the position in which, and at the distance at which, it would be used to swing the airplane. Such movement of the dolly shall not cause a change in compass reading greater than 1/2 degree.

4.4.2 Effect of Movable and Removable Magnetic Parts. The airplane shall be placed on an east or west heading on the ground and the change of compass reading shall be noted while each item is moved through the entire range of movement possible. The change in compass reading shall not exceed 3 degrees. During this procedure the equipment shall be held stationary at the normal position, at the extreme limits of the motion, and at any position where the effect upon the compass is at a maximum, for sufficient time for the compass to settle. Each item shall be tested with all other items in their normal position. This test shall include such items as controls, gun turrents, tail hooks, et cetera. For ammunition, bomb and torpedo loads and other removable items, the compass reading shall be noted with the equipment in place and removed from the airplane. This test shall be repeated on a north or south heading.

4.4.3 Effect of Engine. The airplane shall be placed on an east or west magnetic heading on the ground, the engine or engines shall be run at approximately 1,500 rpm, and the change in compass deviation noted from that with the engines not running. The change in the compass reading shall not exceed 3 degrees. The battery shall not be charging during this test. This test shall be repeated on a north or south heading.

4.4.4 Effect of Variable Circuits. The airplane shall be placed on an east or west magentic heading on the ground, and the change of compass reading produced by the operation of each direct current electrical circuit, which carries a variable current or which intermittently operated during flight, shall be noted. The change in the compass reading shall not exceed 3 degrees. The sum of all individual positive changes due to the effect of variable circuits and the effect of movable and removable magnetic parts shall not exceed 5 degrees for the compass. The sum of all individual negative changes for the compass shall not exceed 5 degrees. This test shall include such circuits as: Generator circuits, including auxiliary power plants, pitot static tube heaters, cockpit and cabin heaters, galley grills, inverters and dynamotors for radio and other equipment, motors for gun turrets and other equipment, cockpit and cabin lights, navigation lights, et cetera. This test may be conducted with the engines either running or not running, at the discretion of the inspector. The generator circuit test may be made with engines running or, if desired, the generator may be replaced by an external source to produce normal generator output current. This test shall be repeated on a north or south heading.

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4.4.5 Effect of Continuously Operated Electrical Circuits. The test procedure of variable circuits shall be followed, except that the tests shall be performed on circuits with draw a steady current. The sum of all individual positive changes caused by electric circuits operated continuously at all times of operation of the airplane shall not exceed 3 degrees for the compass. The sum of all the negative changes for the compass shall not exceed 3 degrees.

4.4.6 Effect of Electrical Circuits, Take-Off and Landing. The test procedure of variable circuits shall be followed, except that the tests shall be run on the circuits for wing flap motors, landing gear and float motors, landing lights, et cetera. The change in compass readings shall not exceed 8 degrees.

4.4.7 Effect of Landing Gear. The effect of landing gear may be tested either on the ground or in flight, at the discretion of the inspector. The change in compass reading shall not exceed 4 degrees.

4.4.7.1 Ground Test. The airplane shall be supported on a north or south magnetic heading by means of suitable jacks or hoists and the compass reading recorded with the landing gear successively in the retracted, lowered, and then retracted position. This procedure shall be repeated on east or west magnetic headings. The apparatus which supports the airplane should preferably be nonmagnetic.

4.4.7.2 Flight Test. The airplane may be flown by the automatic pilot or by reference to a directional gyro on a straight level course on an approximate north magnetic heading with the landing gear retracted. The average compass reading shall be estimataed carefully and recorded. The landing gear shall be lowered and the compass read again. The landing gear shall be retracted and the third compass reading recorded. This procedure shall be repeated with the airplane on an approximate east, south, and west magnetic heading, maximum change of cmpass reading of any heading noted.

4.4.8 Uncompensated Deviation.

4.4.8.1 Ground Swing. With the airplane placed in an east or west heading, the spread between the maximum positive deviation and the maximum negative deviation, with all equipment in its normal position and all electrical circuits turned on, which are normally operated in flight (engine running with battery being charged), shall not be more than 24 degrees for the compass. The procedure shall be repeated for each 30-degree heading throughout a complete revolution.

4.4.8.2 Airswing. The airplane shall be in normal flight conditions and the airswing carried out on 12 approximately evenly spaced magnetic headings. A graph shall be plotted on deviation against the magnetic heading from which the deviation at the 30-degree heading shall be taken. This test may be conducted at the discretion of the inspector. The difference between airswing and ground swing shall not exceed 2 degrees.

4.4.9 Compensated Deviation. The compensator shall be attached to the standard test compass and the compass compensated in accordance with Specification AN-C-106 with the exception that the alinement of the compass on the panel or amount need not be compensated (coefficient "A"). The spread between the maximum positive deviation and the maximum negative deviation shall not exceed 8 degrees for the compass.

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4.5 Rejection and Retest. Rejected installations shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Intended Use. This specification is intended to govern the installation of the standby compass in aircraft

6.2 Definitions.

6.2.1 Deviation. Deviation of an aircraft compass is the difference between the compass heading and the corresponding magnetic heading of the aircraft in which it is installed. Deviation is computed by subtracting the compass reading from the magnetic heading of the aircraft.

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Custodian:
Air Force

Other interest:
Navy - A

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